IAN STALKER

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Summary

Electrical Engineering (Honours) student at UNSW, expected graduation Dec 2027, with hands-on experience across PCB design, analogue circuit design and simulation, 3D CAD and hardware testing. Projects span power electronics, RF receiver design, mixed-signal identification circuits and embedded hardware, taken from simulation through to fabrication and bench verification. Supported by experience in coaching, officiating and event operations.

Technical & Professional Skills

- **PCB & Electronics:** KiCad (schematic capture, layout, fabrication), through-hole and surface-mount soldering, breadboard prototyping
- **Simulation & Analysis:** LTspice, MATLAB (numerical computing, control systems modelling)
- **Test & Measurement:** Oscilloscope analysis, LabVIEW (telecommunications testing)
- **CAD & Fabrication:** Fusion 360, laser cutting
- **Programming:** C, Python, HTML (basic web design), firmware development
- **Professional Skills:** Teamwork and communication, analytical thinking, coaching, attention to detail

Education

Bachelor of Engineering (Honours), Electrical Engineering — Feb 2022 – Dec 2027
University of New South Wales | Sydney, NSW
- Expected graduation: Dec 2027
- WAM of 68

High School Certificate — Feb 2014 – Nov 2021
Hunter Valley Grammar School | Ashtonfield, NSW
- Achieved an ATAR of 93.75

Recent Projects

Electronic Chessboard (87% HD) — 2026
Piece-detecting smart board with a chess opening trainer interface
- Designed a proof-of-concept board using induction coils and passive resonant circuits in each piece to detect both position and identity
- Modelled coil coupling and analogue circuits in LTspice, then designed a custom KiCad PCB (ESP32, USB-C, LDO, ESD protection, LEDs), a Fusion 360 enclosure, and firmware linking to Lichess opening data

ELEC3106 – Fully Analogue Component Identifier (80% D) — 2025
Microcontroller-free open circuit, resistor, capacitor and inductor identifier
- Built a breadboard circuit using a 555 timer, comparators and BJT drivers to classify components across ~510 Ω–49 kΩ, ~150 pF–50 nF and ~5–20 mH, indicated by three LEDs
- Verified against LTspice simulation using an oscilloscope; peak power of 91 mW and total parts cost of $7.50

Buck Converter PCB Design (88% HD) — 2026
MC34063A step-down regulator on a custom two-layer board
- Selected components from the datasheet and laid out a two-layer KiCad PCB, minimising the switching loop and using wide high-current traces and ground pours for EMI and thermal performance
- Assembled the board and verified operation with oscilloscope testing

AM Superheterodyne Receiver Simulation (personal project) — 2026
Full receive chain designed in LTspice
- Designed RF amplifier, switching mixer, two-stage 455 kHz IF filtering, envelope detector and low-pass filter; successfully isolated one station from several simultaneous AM signals and recovered its audio

Work Experience

Assistant Chess Coach
2023 – Present

JZ Master Chess | Carlingford, NSW

- Deliver classroom chess coaching to school-aged students, covering tactical puzzles, game analysis and personalised feedback
- Develop students' strategic thinking, decision-making and overall playing ability

USU Team Member
2022 – 2023

University of Sydney Union | Sydney, NSW

- Supported university events including graduation ceremonies, coordinating staff and communicating with guests, graduates and university officials
- Handled cash, merchandised stock and assisted with daily operations

Football Referee
2017 – 2021

Football NSW Referees Association | Hunter Valley, NSW

- Officiated football matches, enforcing the laws of the game and making impartial decisions on fouls and misconduct
- Managed player behaviour to keep matches fair, safe and compliant with competition regulations

Training & Certificates

Working with Children Check — 2023-present

Referees

Available upon request